

Q1)

```
CREATE DATABASE citymed_hms;
```

```
USE citymed_hms;
```

Statistics 1		
Name	Value	
Updated Rows	1	
Execute time	0.015s	
Start time	Wed May 13 21:04:41 IST 2026	
Finish time	Wed May 13 21:04:41 IST 2026	
Query	-- Q1	
	CREATE DATABASE citymed_hms	

Q2)

```
CREATE TABLE departments(  
  
    dept_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    dept_name VARCHAR(100) NOT NULL unique,  
  
    location VARCHAR(100) NOT NULL,  
  
    phone_ext CHAR(10) NOT NULL,  
  
    head_doctor VARCHAR(50) NULL  
  
);
```

Name	Value
Updated Rows	0
Execute time	0.026s
Start time	Wed May 13 21:07:08 IST 2026
Finish time	Wed May 13 21:07:08 IST 2026
Query	-- Q2
	CREATE TABLE departments(dept_id INT PRIMARY KEY AUTO_INCREMENT, dept_name VARCHAR(100) NOT NULL unique, location VARCHAR(100) NOT NULL, phone_ext CHAR(10) NOT NULL, head_doctor VARCHAR(50) NULL)

Q3)

```
CREATE TABLE wards(

ward_id INT PRIMARY KEY AUTO_INCREMENT,

ward_name VARCHAR(100) NOT NULL,

dept_id INT NOT NULL,

capacity INT NOT NULL check (capacity between 1 and 100),

WARD_TYPE ENUM ('General','ICU','Surgical','Maternity','Paediatric') DEFAULT 'General',

FOREIGN KEY (dept_id) REFERENCES departments(dept_id)

);
```

Name	Value
Updated Rows	0
Execute time	0.035s
Start time	Wed May 13 21:07:44 IST 2026
Finish time	Wed May 13 21:07:45 IST 2026
Query	-- Q3
	CREATE TABLE wards(ward_id INT PRIMARY KEY AUTO_INCREMENT, ward_name VARCHAR(100) NOT NULL, dept_id INT NOT NULL, capacity INT NOT NULL check (capacity between 1 and 100), WARD_TYPE ENUM ('General','ICU','Surgical','Maternity','Paediatric') DEFAULT 'General', FOREIGN KEY (dept_id) REFERENCES departments(dept_id))

Q4)

```
CREATE TABLE doctors(

doctor_id INT PRIMARY KEY AUTO_INCREMENT,
```

```

first_name VARCHAR(100) NOT NULL,

Last_name VARCHAR(100) NOT NULL,

speciality VARCHAR(100) NOT NULL,

dep_id INT NOT NULL,

email VARCHAR(100) UNIQUE,

phone INT NOT NULL,

hir_date DATE NOT NULL,

salary DECIMAL(10,2) NOT NULL CHECK(salary > 0),

FOREIGN KEY (dep_id) REFERENCES departments(dept_id)

);

```

Name	Value
Updated Rows	0
Execute time	0.023s
Start time	Wed May 13 21:08:49 IST 2026
Finish time	Wed May 13 21:08:49 IST 2026
Query	-- Q4
	CREATE TABLE doctors(
	doctor_id INT PRIMARY KEY AUTO_INCREMENT,
	first_name VARCHAR(100) NOT NULL,
	Last_name VARCHAR(100) NOT NULL,
	speciality VARCHAR(100) NOT NULL,
	dep_id INT NOT NULL,
	email VARCHAR(100) UNIQUE,
	phone INT NOT NULL,
	hir_date DATE NOT NULL,
	salary DECIMAL(10,2) NOT NULL CHECK(salary > 0),
	FOREIGN KEY (dep_id) REFERENCES departments(dept_id)
	)

Q5)

```

CREATE TABLE nurses(

nurse_id INT PRIMARY KEY AUTO_INCREMENT,

first_name VARCHAR(100) NOT NULL,

last_name VARCHAR(100) NOT NULL,

ward_id INT NOT NULL,

shift ENUM ('Morning','Evening','Night') NOT NULL,

email VARCHAR(50) NOT NULL unique,

FOREIGN KEY (ward_id) REFERENCES wards(ward_id)

```

);

Statistics 1	
Name	Value
Updated Rows	0
Execute time	0.026s
Start time	Wed May 13 21:09:52 IST 2026
Finish time	Wed May 13 21:09:52 IST 2026
Query	-- Q5
	CREATE TABLE nurses(nurse_id INT PRIMARY KEY AUTO_INCREMENT, first_name VARCHAR(100) NOT NULL, last_name VARCHAR(100) NOT NULL, ward_id INT NOT NULL, shift ENUM ('Morning','Evening','Night') NOT NULL, email VARCHAR(50) NOT NULL unique, FOREIGN KEY (ward_id) REFERENCES wards(ward_id))

Q6)

CREATE TABLE patients(
patient_id **INT PRIMARY KEY AUTO_INCREMENT**,

first_name **VARCHAR(100) NOT NULL**,

last_name **VARCHAR(100) NOT NULL**,

dob **DATE NOT NULL**,

gender **ENUM('Male','Female','Other') NOT NULL**,

blood_group **ENUM('A+','A-','B+','B-','AB+','O+','O-') NULL**,

phone **VARCHAR(20) NOT NULL**,

email **VARCHAR(50) NULL**,

address **TEXT NULL**,

registered_on **DATE NOT NULL DEFAULT (CURRENT_DATE)**

);

Name	Value
Updated Rows	0
Execute time	0.019s
Start time	Wed May 13 21:10:32 IST 2026
Finish time	Wed May 13 21:10:32 IST 2026
Query	-- Q6
	CREATE TABLE patients(patient_id INT PRIMARY KEY AUTO_INCREMENT, first_name VARCHAR(100) NOT NULL, last_name VARCHAR(100) NOT NULL, dob DATE NOT NULL, gender ENUM('Male','Female','Other') NOT NULL, blood_group ENUM('A+','A-','B+','B-','AB+','O+','O-') NULL, phone VARCHAR(20) NOT NULL, email VARCHAR(50) NULL, address TEXT NULL, registered_on DATE NOT NULL DEFAULT (CURRENT_DATE))

Q7)

CREATE TABLE appointments(

appointment_id **INT PRIMARY KEY AUTO_INCREMENT**,

patient_id **INT NOT NULL**,

doctor_id **INT NOT NULL**,

appt_date **DATE NOT NULL**,

appt_time **TIME NOT NULL**,

reason **VARCHAR(255) NOT NULL**,

status **ENUM('Scheduled','Completed','Cancelled','No-Show') DEFAULT 'Scheduled'**,

notes **TEXT NULL**,

FOREIGN KEY (doctor_id) **REFERENCES** doctors(doctor_id),

FOREIGN KEY (patient_id) **REFERENCES** patients(patient_id)

);

Name	Value
Updated Rows	0
Execute time	0.027s
Start time	Wed May 13 21:11:39 IST 2026
Finish time	Wed May 13 21:11:39 IST 2026
Query	-- Q7
	CREATE TABLE appointments(appointment_id INT PRIMARY KEY AUTO_INCREMENT, patient_id INT NOT NULL, doctor_id INT NOT NULL, appt_date DATE NOT NULL, appt_time TIME NOT NULL, reason VARCHAR(255) NOT NULL, status ENUM('Scheduled','Completed','Cancelled','No-Show') DEFAULT 'Scheduled', notes TEXT NULL, FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id), FOREIGN KEY (patient_id) REFERENCES patients(patient_id))

Q8)

```
CREATE TABLE prescriptions(  
  
    prescription_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    appointment_id INT NOT NULL,  
  
    medicine_name VARCHAR(150) NOT NULL,  
  
    dosage VARCHAR(50) NOT NULL,  
  
    duration_days INT UNSIGNED NOT NULL CHECK(duration_days BETWEEN 1 and  
365),  
  
    instructions TEXT NULL,  
  
    issued_date DATE NOT NULL DEFAULT(CURRENT_DATE),  
  
    FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id)  
  
        ON DELETE CASCADE  
  
        ON UPDATE CASCADE,  
  
    UNIQUE KEY unique_medicine_per_appointment (appointment_id, medicine_name)  
  
);
```

Statistics 1	
Name	Value
Updated Rows	0
Execute time	0.02s
Start time	Wed May 13 21:12:21 IST 2026
Finish time	Wed May 13 21:12:21 IST 2026
Query	-- Q8 CREATE TABLE prescriptions(prescription_id INT PRIMARY KEY AUTO_INCREMENT, appointment_id INT NOT NULL, medicine_name VARCHAR(150) NOT NULL, dosage VARCHAR(50) NOT NULL, duration_days INT UNSIGNED NOT NULL CHECK(duration_days BETWEEN 1 and 365), instructions TEXT NULL, issued_date DATE NOT NULL DEFAULT(CURRENT_DATE), FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id) ON DELETE CASCADE ON UPDATE CASCADE, UNIQUE KEY unique_medicine_per_appointment (appointment_id, medicine_name))

Q9)

```
CREATE TABLE bills(  
  
    bill_id INT PRIMARY KEY AUTO_INCREMENT,
```

```

appointment_id INT NOT NULL UNIQUE,

totle_amount DECIMAL(10,2) NOT NULL CHECK(totle_amount >=0),

paid_amount DECIMAL(10,2) NOT NULL DEFAULT '0.00' CHECK(paid_amount
>=0),

payment_ststus ENUM('Unpaid','Partial','Paid') DEFAULT 'Unpaid',

billing_date DATE NOT NULL DEFAULT(CURRENT_DATE),

FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id),

CHECK(paid_amount <= totle_amount)

);

```

Name	Value
Updated Rows	0
Execute time	0.024s
Start time	Wed May 13 21:21:43 IST 2026
Finish time	Wed May 13 21:21:43 IST 2026
Query	-- Q9
	CREATE TABLE bills(bill_id INT PRIMARY KEY AUTO_INCREMENT, appointment_id INT NOT NULL UNIQUE, totle_amount DECIMAL(10,2) NOT NULL CHECK(totle_amount >=0), paid_amount DECIMAL(10,2) NOT NULL DEFAULT '0.00' CHECK(paid_amount >=0), payment_ststus ENUM('Unpaid','Partial','Paid') DEFAULT 'Unpaid', billing_date DATE NOT NULL DEFAULT(CURRENT_DATE), FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id), CHECK(paid_amount <= totle_amount))

Q10)

ALTER TABLE patients

ADD emergency_contact **VARCHAR**(50) **null**;

Name	Value
Updated Rows	0
Execute time	0.02s
Start time	Wed May 13 21:30:32 IST 2026
Finish time	Wed May 13 21:30:32 IST 2026
Query	-- Q10
	ALTER TABLE patients
	ADD emergency_contact VARCHAR(50) null

Q11)

ALTER TABLE doctors

MODIFY phone **VARCHAR** (50) **not null**;

Statistics 1	
Name	Value
Updated Rows	0
Execute time	0.06s
Start time	Wed May 13 23:19:30 IST 2026
Finish time	Wed May 13 23:19:30 IST 2026
Query	-- Q11
	ALTER TABLE doctors
	MODIFY phone VARCHAR (50) not null

Q12)

ALTER TABLE appointments

ADD CONSTRAINT chk_appt_date_future

CHECK (appt_date >= **CURRENT_DATE**);

- All tables

SHOW TABLES;

Results 1	
SHOW TABLES Enter a SQL expression to f	
Grid	AZ Tables_in_citymed_hms
1	appointments
2	bills
3	departments
4	doctors
5	nurses
6	patients
7	prescriptions
8	wards

DESC departments;

columns 1 X							
DESC departments							
Connection: localhost Time: 2026-05-14 00:05:07.737 Query: DESC departments							
	A-Z Field	A-Z Type	A-Z Null	A-Z Key	A-Z Default	A-Z Extra	
1	dept_id	int	NO	PRI	[NULL]	auto_increment	
2	dept_name	varchar(100)	NO	UNI	[NULL]		
3	location	varchar(100)	NO		[NULL]		
4	phone_ext	char(10)	NO		[NULL]		
5	head_doctor	varchar(50)	YES		[NULL]		

DESC wards;

columns 1 X							
DESC wards							
	A-Z Field	A-Z Type	A-Z Null	A-Z Key	A-Z Default	A-Z Extra	
1	ward_id	int	NO	PRI	[NULL]	auto_increment	
2	ward_name	varchar(100)	NO		[NULL]		
3	dept_id	int	NO	MUL	[NULL]		
4	capacity	int	NO		[NULL]		
5	WARD_TYPE	enum('General','IC	YES		General		

DESC doctors;

columns 1 X							
DESC doctors							
Enter a SQL expression to filter results (use Ctrl+Space)							
	A-Z Field	A-Z Type	A-Z Null	A-Z Key	A-Z Default	A-Z Extra	
1	doctor_id	int	NO	PRI	[NULL]	auto_increment	
2	first_name	varchar(100)	NO		[NULL]		
3	Last_name	varchar(100)	NO		[NULL]		
4	speciality	varchar(100)	NO		[NULL]		
5	dep_id	int	NO	MUL	[NULL]		
6	email	varchar(100)	YES	UNI	[NULL]		
7	phone	varchar(50)	NO		[NULL]		
8	hir_date	date	NO		[NULL]		
9	salary	decimal(10,2)	NO		[NULL]		

DESC nurses;

columns 1		DESC nurses Enter a SQL expression to filter results (use Ctrl+Space)					
	AZ Field	AZ Type	AZ Null	AZ Key	AZ Default	AZ Extra	
1	nurse_id	int	NO	PRI	[NULL]	auto_increment	
2	first_name	varchar(100)	NO		[NULL]		
3	last_name	varchar(100)	NO		[NULL]		
4	ward_id	int	NO	MUL	[NULL]		
5	shift	enum('Morning','E	NO		[NULL]		
6	email	varchar(50)	NO	UNI	[NULL]		

DESC patients;

columns 1		DESC patients Enter a SQL expression to filter results (use Ctrl+Space)					
	AZ Field	AZ Type	AZ Null	AZ Key	AZ Default	AZ Extra	
1	patient_id	int	NO	PRI	[NULL]	auto_increment	
2	first_name	varchar(100)	NO		[NULL]		
3	last_name	varchar(100)	NO		[NULL]		
4	dob	date	NO		[NULL]		
5	gender	enum('Male','Fem	NO		[NULL]		
6	blood_group	enum('A+','A-','B+	YES		[NULL]		
7	phone	varchar(20)	NO		[NULL]		
8	email	varchar(50)	YES		[NULL]		
9	address	text	YES		[NULL]		
10	registered_on	date	NO		curdate()	DEFAULT_GENER.	
11	emergency_conta	varchar(50)	YES		[NULL]		

DESC appointments;

columns 1		DESC appointments Enter a SQL expression to filter results (use Ctrl+Space)					
	AZ Field	AZ Type	AZ Null	AZ Key	AZ Default	AZ Extra	
1	appointment_id	int	NO	PRI	[NULL]	auto_increment	
2	patient_id	int	NO	MUL	[NULL]		
3	doctor_id	int	NO	MUL	[NULL]		
4	appt_date	date	NO		[NULL]		
5	appt_time	time	NO		[NULL]		
6	reason	varchar(255)	NO		[NULL]		
7	status	enum('Scheduled	YES		Scheduled		
8	notes	text	YES		[NULL]		

DESC prescriptions;

columns 1 X							
DESC prescriptions Enter a SQL expression to filter results (use Ctrl+Space)							
	A-Z Field	A-Z Type	A-Z Null	A-Z Key	A-Z Default	A-Z Extra	
1	prescription_id	int	NO	PRI	[NULL]	auto_increment	
2	appointment_id	int	NO	MUL	[NULL]		
3	medicine_name	varchar(150)	NO		[NULL]		
4	dosage	varchar(50)	NO		[NULL]		
5	duration_days	int unsigned	NO		[NULL]		
6	instructions	text	YES		[NULL]		
7	issued_date	date	NO		curdate()	DEFAULT_GENER	

DESC bills;

columns 1 X							
DESC bills Enter a SQL expression to filter results (use Ctrl+Space)							
	A-Z Field	A-Z Type	A-Z Null	A-Z Key	A-Z Default	A-Z Extra	
1	bill_id	int	NO	PRI	[NULL]	auto_increment	
2	appointment_id	int	NO	UNI	[NULL]		
3	totle_amount	decimal(10,2)	NO		[NULL]		
4	paid_amount	decimal(10,2)	NO		0.00		
5	payment_ststus	enum('Unpaid','Pa	YES		Unpaid		
6	billing_date	date	NO		curdate()	DEFAULT_GENER	

Q13)

DROP TABLE IF EXISTS bills;

Statistics 1 X	
Name	Value
Updated Rows	0
Execute time	0.031s
Start time	Thu May 14 00:38:20 IST 2026
Finish time	Thu May 14 00:38:20 IST 2026
Query	-- Q13
	DROP TABLE IF EXISTS bills

DROP TABLE IF EXISTS prescriptions;

Statistics 1 X		Output
Name	Value	
Updated Rows	0	
Execute time	0.02s	
Start time	Thu May 14 00:47:41 IST 2026	
Finish time	Thu May 14 00:47:41 IST 2026	
Query	DROP TABLE IF EXISTS prescriptions	

DROP TABLE IF EXISTS appointments;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.026s
Start time	Thu May 14 00:48:41 IST 2026
Finish time	Thu May 14 00:48:41 IST 2026
Query	DROP TABLE IF EXISTS appointments

DROP TABLE IF EXISTS nurses;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.017s
Start time	Thu May 14 00:49:27 IST 2026
Finish time	Thu May 14 00:49:27 IST 2026
Query	DROP TABLE IF EXISTS nurses

DROP TABLE IF EXISTS wards;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.019s
Start time	Thu May 14 00:50:09 IST 2026
Finish time	Thu May 14 00:50:09 IST 2026
Query	DROP TABLE IF EXISTS wards

DROP TABLE IF EXISTS doctoes;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.006s
Start time	Thu May 14 00:50:40 IST 2026
Finish time	Thu May 14 00:50:40 IST 2026
Query	DROP TABLE IF EXISTS doctoes

DROP TABLE IF EXISTS patient;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.004s
Start time	Thu May 14 00:51:18 IST 2026
Finish time	Thu May 14 00:51:18 IST 2026
Query	DROP TABLE IF EXISTS patient

DROP TABLE IF EXISTS departments;

Statistics 1 × Output	
Name	Value
Updated Rows	0
Execute time	0.003s
Start time	Thu May 14 00:51:51 IST 2026
Finish time	Thu May 14 00:51:51 IST 2026
Query	DROP TABLE IF EXISTS departments

-- පැහැදිලි කිරීම: මෙම අනුපිළිවෙල අනුගමනය කරන්නේ foreign key constraints නිසාය.

-- Child tables (foreign keys ඇති tables) parent tables වලට පෙර drop කළ යුතුය.

-- එසේ නොමැතිනම් foreign key constraint violation error එකක් ඇතිවේ.

Q14)

INSERT INTO departments (dept_name, location, phone_ext, head_doctor)

VALUES

('Cardiology', '3rd Floor, East Wing', '1234', 'Dr. John Smith'),

('Orthopaedics', '2nd Floor, West Wing', '5678', 'Dr. Sarah Lee'),

('Neurology', '4th Floor, North Wing', '9012', 'Dr. Michael Brown'),

('Paediatrics', '1st Floor, South Wing', '3456', 'Dr. Emily White');

SELECT * FROM departments;

departments 1 × Output	
SELECT * FROM departments Enter a SQL expression to filter results (use Ctrl+Space)	
Grid	123 dept_id AZ dept_name AZ location AZ phone_ext AZ head_doctor
1	1 Cardiology 3rd Floor, East Wing 1234 Dr. John Smith
2	2 Orthopaedics 2nd Floor, West Wing 5678 Dr. Sarah Lee
3	3 Neurology 4th Floor, North Wing 9012 Dr. Michael Brown
4	4 Paediatrics 1st Floor, South Wing 3456 Dr. Emily White

Q15)

INSERT INTO wards (ward_name, dept_id, capacity, ward_type)

VALUES

```
('Cardiac Care ICU', 1, 25, 'ICU'),  
('Cardiology General Ward', 1, 40, 'General'),  
('Orthopaedic Surgical Ward', 2, 30, 'Surgical'),  
('Orthopaedic Recovery Ward', 2, 35, 'General'),  
('Neurology ICU', 3, 20, 'ICU'),  
('Neurology General Ward', 3, 45, 'General'),  
('Paediatric Ward', 4, 50, 'Paediatric'),  
('Neonatal ICU', 4, 15, 'ICU');
```

SELECT * FROM wards;

wards 1

Output

SELECT * FROM wards

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

Text

	123 ward_id	A-Z ward_name	123 dept_id	123 capacity	A-Z ward_type
1	1	Cardiac Care ICU	1	25	ICU
2	2	Cardiology General Ward	1	40	General
3	3	Orthopaedic Surgical Ward	2	30	Surgical
4	4	Orthopaedic Recovery Ward	2	35	General
5	5	Neurology ICU	3	20	ICU
6	6	Neurology General Ward	3	45	General
7	7	Paediatric Ward	4	50	Paediatric
8	8	Neonatal ICU	4	15	ICU

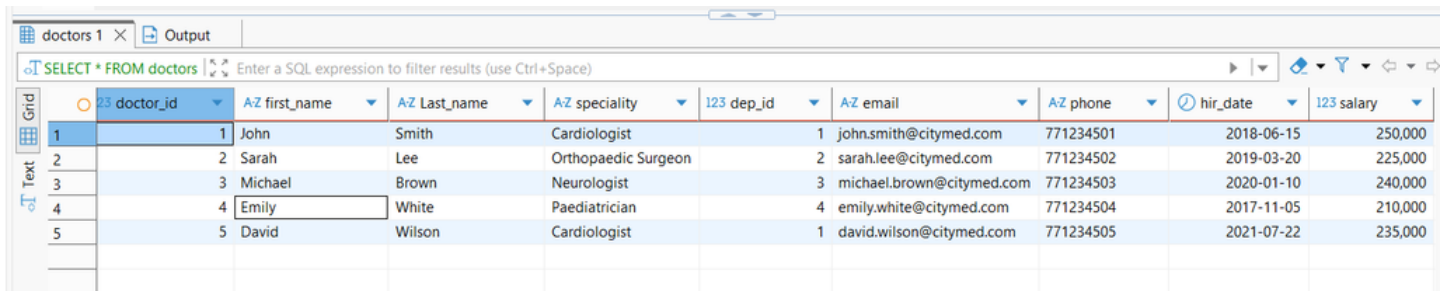
Q16)

INSERT INTO doctors (first_name, Last_name, speciality, dep_id, email, phone, hir_date, salary)

VALUES

```
('John', 'Smith', 'Cardiologist', 1, 'john.smith@citymed.com', 771234501, '2018-06-15', 250000.00),  
('Sarah', 'Lee', 'Orthopaedic Surgeon', 2, 'sarah.lee@citymed.com', 771234502, '2019-03-20', 225000.00),  
('Michael', 'Brown', 'Neurologist', 3, 'michael.brown@citymed.com', 771234503, '2020-01-10', 240000.00),  
('Emily', 'White', 'Paediatrician', 4, 'emily.white@citymed.com', 771234504, '2017-11-05', 210000.00),  
('David', 'Wilson', 'Cardiologist', 1, 'david.wilson@citymed.com', 771234505, '2021-07-22', 235000.00);
```

SELECT * FROM doctors;



	123 doctor_id	AZ first_name	AZ Last_name	AZ speciality	123 dep_id	AZ email	AZ phone	hir_date	123 salary
1	1	John	Smith	Cardiologist	1	john.smith@citymed.com	771234501	2018-06-15	250,000
2	2	Sarah	Lee	Orthopaedic Surgeon	2	sarah.lee@citymed.com	771234502	2019-03-20	225,000
3	3	Michael	Brown	Neurologist	3	michael.brown@citymed.com	771234503	2020-01-10	240,000
4	4	Emily	White	Paediatrician	4	emily.white@citymed.com	771234504	2017-11-05	210,000
5	5	David	Wilson	Cardiologist	1	david.wilson@citymed.com	771234505	2021-07-22	235,000

Q17)

INSERT INTO nurses (first_name, last_name, ward_id, shift, email)

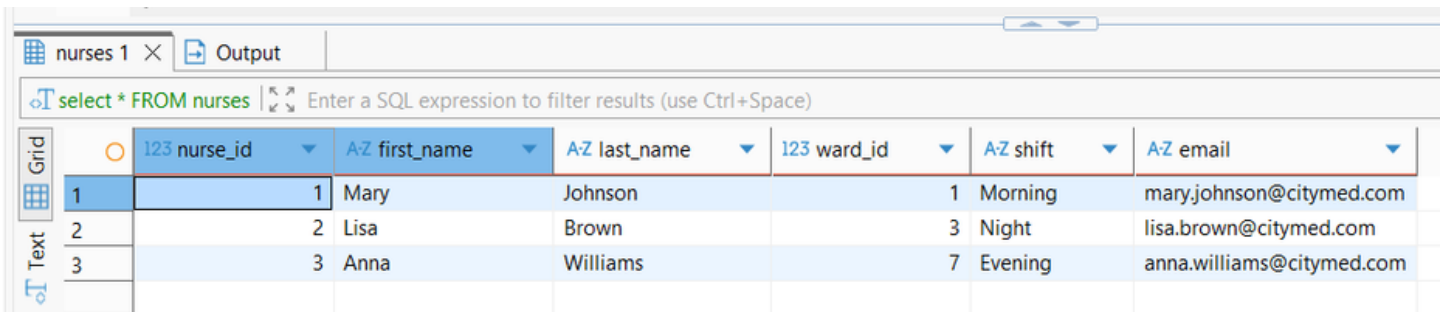
VALUES

('Mary', 'Johnson', 1, 'Morning', 'mary.johnson@citymed.com'),

('Lisa', 'Brown', 3, 'Night', 'lisa.brown@citymed.com'),

('Anna', 'Williams', 7, 'Evening', 'anna.williams@citymed.com');

select * FROM nurses;



	123 nurse_id	AZ first_name	AZ last_name	123 ward_id	AZ shift	AZ email
1	1	Mary	Johnson	1	Morning	mary.johnson@citymed.com
2	2	Lisa	Brown	3	Night	lisa.brown@citymed.com
3	3	Anna	Williams	7	Evening	anna.williams@citymed.com

Q18)

INSERT INTO patients (first_name, last_name, dob, gender, blood_group, phone, email, address)

VALUES

('Kamal', 'Perera', '1985-05-15', 'Male', 'O+', '0712345678', 'kamal.perera@email.com', 'Colombo'),

('Nimali', 'Silva', '1990-10-20', 'Female', 'A-', '0723456789', 'nimali.silva@email.com', 'Kandy'),

('Sunil', 'Jayawardena', '1978-03-12', 'Male', 'B+', '0734567890', **NULL**, 'Galle'),

('Amali', 'Fernando', '1995-07-25', 'Female', 'AB+', '0745678901', 'amali.f@email.com', 'Negombo'),

('Ruwan', 'Rathnayake', '2000-12-01', 'Other', 'O-', '0756789012', 'ruwan.r@email.com', 'Kurunegala'),
('Dilani', 'Bandara', '1988-08-18', 'Female', 'A+', '0767890123', NULL, 'Matara');

SELECT * FROM patients;

patients 1 × Output										
SELECT * FROM patients Enter a SQL expression to filter results (use Ctrl+Space)										
Grid	123 patient_id	AZ first_name	AZ last_name	dob	AZ gender	AZ blood_group	AZ phone	AZ email	AZ address	
1	1	Kamal	Perera	1985-05-15	Male	O+	0712345678	kamal.perera@email.com	Colombo	
2	2	Nimali	Silva	1990-10-20	Female	A-	0723456789	nimali.silva@email.com	Kandy	
3	3	Sunil	Jayawardena	1978-03-12	Male	B+	0734567890	[NULL]	Galle	
4	4	Amali	Fernando	1995-07-25	Female	AB+	0745678901	amali.f@email.com	Negombo	
5	5	Ruwan	Rathnayake	2000-12-01	Other	O-	0756789012	ruwan.r@email.com	Kurunegala	
6	6	Dilani	Bandara	1988-08-18	Female	A+	0767890123	[NULL]	Matara	

Q19)

INSERT INTO appointments (patient_id, doctor_id, appt_date, appt_time, reason, status, notes)

VALUES

(1, 1, '2024-01-10', '09:00:00', 'Chest pain', 'Completed', 'ECG normal'),
(2, 2, '2024-01-12', '10:30:00', 'Knee pain', 'Completed', 'Physiotherapy'),
(3, 3, '2024-01-15', '14:00:00', 'Headache', 'Completed', 'MRI scheduled'),
(4, 4, '2024-01-18', '11:00:00', 'Vaccination', 'Completed', 'Done'),
(1, 3, '2024-02-01', '09:30:00', 'Neurology check', 'Scheduled', NULL),
(2, 1, '2024-02-03', '13:00:00', 'Cardiac check', 'Scheduled', NULL),
(5, 2, '2024-02-05', '10:00:00', 'Back pain', 'Scheduled', NULL),
(3, 1, '2024-01-05', '08:30:00', 'Checkup', 'Cancelled', 'Patient cancelled');

SELECT * FROM appointments;

appointments 1 × Output										
SELECT * FROM appointments Enter a SQL expression to filter results (use Ctrl+Space)										
Grid	123 appointment_id	123 patient_id	123 doctor_id	appt_date	appt_time	AZ reason	AZ status	AZ notes		
1	1	1	1	2024-01-10	09:00:00	Chest pain	Completed	ECG normal		
2	2	2	2	2024-01-12	10:30:00	Knee pain	Completed	Physiotherapy		
3	3	3	3	2024-01-15	14:00:00	Headache	Completed	MRI scheduled		
4	4	4	4	2024-01-18	11:00:00	Vaccination	Completed	Done		
5	5	1	3	2024-02-01	09:30:00	Neurology check	Scheduled	[NULL]		
6	6	2	1	2024-02-03	13:00:00	Cardiac check	Scheduled	[NULL]		
7	7	5	2	2024-02-05	10:00:00	Back pain	Scheduled	[NULL]		
8	8	3	1	2024-01-05	08:30:00	Checkup	Cancelled	Patient cancelled		

Q20)

INSERT INTO prescriptions (appointment_id, medicine_name, dosage, duration_days, instructions)

VALUES

```
(1, 'Aspirin', '75mg once daily', 30, 'Take after breakfast'),
(1, 'Atorvastatin', '20mg once daily', 90, 'Take at bedtime'),
(2, 'Ibuprofen', '400mg three times daily', 7, 'Take after meals'),
(3, 'Paracetamol', '500mg twice daily', 5, 'Take when headache occurs'),
(4, 'Vitamin C', '500mg once daily', 30, 'Take after breakfast');
```

```
SELECT * FROM prescriptions;
```

prescriptions 1 × Output

SELECT * FROM prescriptions

	123 prescription_id	123 appointment_id	AZ medicine_name	AZ dosage	123 duration_days	AZ instructions	issued_date
1	1	1	Aspirin	75mg once daily	30	Take after breakfast	2026-05-14
2	2	1	Atorvastatin	20mg once daily	90	Take at bedtime	2026-05-14
3	3	2	Ibuprofen	400mg three times daily	7	Take after meals	2026-05-14
4	4	3	Paracetamol	500mg twice daily	5	Take when headache occurs	2026-05-14
5	5	4	Vitamin C	500mg once daily	30	Take after breakfast	2026-05-14

Q21)

```
INSERT INTO bills (appointment_id, totle_amount, paid_amount, payment_ststus, billing_date)
```

VALUES

```
(1, 7500.00, 7500.00, 'Paid', '2024-01-10'),  
(2, 5000.00, 2000.00, 'Partial', '2024-01-12'),  
(3, 3500.00, 0.00, 'Unpaid', '2024-01-15'),  
(4, 2500.00, 2500.00, 'Paid', '2024-01-18');
```

```
SELECT * FROM bills;
```

bills 1 ×
Output

SELECT * FROM bills
Enter a SQL expression to filter results (use Ctrl+Space)

	123 bill_id	123 appointment_id	123 totle_amount	123 paid_amount	A-Z payment_ststus	🕒 billing_date
1	1	1	7,500	7,500	Paid	2024-01-10
2	2	2	5,000	2,000	Partial	2024-01-12
3	3	3	3,500	0	Unpaid	2024-01-15
4	4	4	2,500	2,500	Paid	2024-01-18

Q22)

UPDATE appointments

SET status = 'Cancelled',

notes = 'Appointment cancelled by patient due to personal reasons'

WHERE appointment_id = 8;

SELECT * FROM appointments;

appointments 1 ×

Output

SELECT * FROM appointments

Enter a SQL expression to filter results (use Ctrl+Space)

	123 appointment_id	123 patient_id	123 doctor_id	appt_date	appt_time	AZ reason	AZ status	AZ notes
1	1	1	1	2024-01-10	09:00:00	Chest pain	Completed	ECG normal
2	2	2	2	2024-01-12	10:30:00	Knee pain	Completed	Physiotherapy
3	3	3	3	2024-01-15	14:00:00	Headache	Completed	MRI scheduled
4	4	4	4	2024-01-18	11:00:00	Vaccination	Completed	Done
5	5	5	1	2024-02-01	09:30:00	Neurology check	Scheduled	[NULL]
6	6	6	2	2024-02-03	13:00:00	Cardiac check	Scheduled	[NULL]
7	7	7	5	2024-02-05	10:00:00	Back pain	Scheduled	[NULL]
8	8	8	3	2024-01-05	08:30:00	Checkup	Cancelled	Appointment cancelled by patient due to

Q23)

UPDATE bills

SET paid_amount = LEAST(paid_amount + 5000, totle_amount),

payment_ststatus = CASE

WHEN (paid_amount + 5000) >= totle_amount

THEN 'Paid'

ELSE 'Partial'

END

WHERE bill_id = 2;

SELECT * FROM bills;

bills 1

×

SELECT * FROM bills

↕ ↕

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

Text

	123 bill_id	123 appointment	123 totle_amount	123 paid_amount	A-Z payment_ststu	🕒 billing_date
1	1	1	7,500	7,500	Paid	2024-01-10
2	2	2	5,000	5,000	Paid	2024-01-12
3	3	3	3,500	0	Unpaid	2024-01-15
4	4	4	2,500	2,500	Paid	2024-01-18

Q24)

UPDATE doctors

JOIN departments ON doctors.dep_id = departments.dept_id

set salary = salary * 0.10

WHERE departments.dept_name = 'Cardiologist';

SELECT * FROM doctors;

doctors 1 X									
SELECT * FROM doctors Enter a SQL expression to filter results (use Ctrl+Space)									
	123 doctor_id	AZ first_name	AZ Last_name	AZ speciality	123 dep_id	AZ email	AZ phone	hir_date	123 salary
1	1	John	Smith	Cardiologist	1	john.smith@citymed.c	771234501	2018-06-15	250,000
2	2	Sarah	Lee	Orthopaedic Surgeon	2	sarah.lee@citymed.co	771234502	2019-03-20	225,000
3	3	Michael	Brown	Neurologist	3	michael.brown@citym	771234503	2020-01-10	240,000
4	4	Emily	White	Paediatrician	4	emily.white@citymed.	771234504	2017-11-05	210,000
5	5	David	Wilson	Cardiologist	1	david.wilson@citymed	771234505	2021-07-22	235,000

Q25)

DELETE FROM bills

WHERE appointment_id in (

SELECT appointment_id

FROM appointments

WHERE status = 'cancelled'

);

SELECT * FROM bills;

bills 1 X							
SELECT * FROM bills Enter a SQL expression to filter results (use Ctrl+Space)							
	123 bill_id	123 appointment_id	123 totle_amount	123 paid_amount	AZ payment_ststus	billing_date	
1	1	1	7,500	7,500	Paid	2024-01-10	
2	2	2	5,000	5,000	Paid	2024-01-12	
3	3	3	3,500	0	Unpaid	2024-01-15	
4	4	4	2,500	2,500	Paid	2024-01-18	

Q26)

DELETE FROM appointments

```
WHERE status = 'No Show'

AND appt_date < DATE_SUB(CURDATE(), INTERVAL 2 YEAR);
```

```
SELECT * FROM appointments;
```

appointments 1									
SELECT * FROM appointments									
	123 appointment_id	123 patient_id	123 doctor_id	appt_date	appt_time	AZ reason	AZ status	AZ notes	
1	1	1	1	2024-01-10	09:00:00	Chest pain	Completed	ECG normal	
2	2	2	2	2024-01-12	10:30:00	Knee pain	Completed	Physiotherapy	
3	3	3	3	2024-01-15	14:00:00	Headache	Completed	MRI scheduled	
4	4	4	4	2024-01-18	11:00:00	Vaccination	Completed	Done	
5	5	1	3	2024-02-01	09:30:00	Neurology check	Scheduled	[NULL]	
6	6	2	1	2024-02-03	13:00:00	Cardiac check	Scheduled	[NULL]	
7	7	5	2	2024-02-05	10:00:00	Back pain	Scheduled	[NULL]	
8	8	3	1	2024-01-05	08:30:00	Checkup	Cancelled	Appointment cancelled	

Q27)

```
SELECT
```

```
patient_id,

CONCAT(first_name,"last_name)as full_name,

dob,

gender,

blood_group,

phone
```

```
FROM patients
```

```
ORDER BY last_name;
```

patients 1							
SELECT patient_id, CONCAT(first_name,"last_name)							
	123 patient_id	AZ full_name	dob	AZ gender	AZ blood_group	AZ phone	
1	6	DilaniBandara	1988-08-18	Female	A+	0767890123	
2	4	AmaliFernando	1995-07-25	Female	AB+	0745678901	
3	3	SunilJayawardena	1978-03-12	Male	B+	0734567890	
4	1	KamalPerera	1985-05-15	Male	O+	0712345678	
5	5	RuwanRathnayake	2000-12-01	Other	O-	0756789012	
6	2	NimaliSilva	1990-10-20	Female	A-	0723456789	

Q28)

select

CONCAT (patients.first_name,',patients.last_name)**as** *patient_name*,

CONCAT (doctors.first_name,',doctors.Last_name)**as** *doctor_name*,

appt_date,

appt_time,

reason

from appointments

inner join patients **on** appointments.patient_id = patients.patient_id

inner join doctors **on** appointments.doctor_id = doctors.doctor_id

where appointments.status = 'Scheduled' **and** appointments.appt_date >= **CURDATE**()

order by appointments.appt_date,appointments.appt_time;

The screenshot shows a database query editor window titled 'appointments 1'. The SQL query entered is: `SELECT CONCAT(p.first_name, ', ', p.last_name) AS p`. Below the query bar is a result grid with 6 columns: 'A-Z patient_name', 'A-Z doctor_name', 'appt_date', 'appt_time', and 'A-Z reason'. The grid is currently empty, showing only the header row.

A-Z patient_name	A-Z doctor_name	appt_date	appt_time	A-Z reason

Today's date (CURDATE()) = 2026-05-16 (the current year is 2026)

Since all these appt_date values are in 2024, they are earlier than (past) CURDATE().

Therefore: no rows will match the condition appt_date >= CURDATE().

Q29)

SELECT

appointments.appointment_id,

CONCAT(patients.first_name, ', ', patients.last_name) **AS** *patient_name*,

CONCAT(doctors.first_name, ', ', doctors.last_name) **AS** *doctor_name*,

departments.dept_name **AS** *department_name*,

appointments.appt_date,

appointments.appt_time,

appointments.status

FROM appointments

INNER JOIN patients **ON** appointments.patient_id = patients.patient_id

INNER JOIN doctors **ON** appointments.doctor_id = doctors.doctor_id

INNER JOIN departments **ON** doctors.dep_id = departments.dept_id;

appointments(+) 1 X							
SELECT appointments.appointment_id, CONCAT(p, Enter a SQL expression to filter results (use Ctrl+Space)							
Grid	123 appointment_id	AZ patient_name	AZ doctor_name	AZ department_name	appt_date	appt_time	AZ status
1	1	Kamal Perera	John Smith	Cardiology	2024-01-10	09:00:00	Completed
2	6	Nimali Silva	John Smith	Cardiology	2024-02-03	13:00:00	Scheduled
3	8	Sunil Jayawardena	John Smith	Cardiology	2024-01-05	08:30:00	Cancelled
4	3	Sunil Jayawardena	Michael Brown	Neurology	2024-01-15	14:00:00	Completed
5	5	Kamal Perera	Michael Brown	Neurology	2024-02-01	09:30:00	Scheduled
6	2	Nimali Silva	Sarah Lee	Orthopaedics	2024-01-12	10:30:00	Completed
7	7	Ruwan Rathnayake	Sarah Lee	Orthopaedics	2024-02-05	10:00:00	Scheduled
8	4	Amali Fernando	Emily White	Paediatrics	2024-01-18	11:00:00	Completed

Q30)

SELECT

prescriptions.prescription_id,

CONCAT(patients.first_name, ' ', patients.last_name) **AS** patient_name,

CONCAT(doctors.first_name, ' ', doctors.last_name) **AS** doctor_name,

prescriptions.medicine_name,

prescriptions.dosage,

prescriptions.duration_days **AS** duration

FROM prescriptions

INNER JOIN appointments **ON** prescriptions.appointment_id = appointments.appointment_id

INNER JOIN patients **ON** appointments.patient_id = patients.patient_id

INNER JOIN doctors **ON** appointments.doctor_id = doctors.doctor_id

where appointments.status = 'Completed';

prescriptions 1 X						
SELECT prescriptions.prescri Enter a SQL expression to filter results (use Ctrl+Space)						
	123 prescription_id	AZ patient_name	AZ doctor_name	AZ medicine_name	AZ dosage	123 duration
1	1	Kamal Perera	John Smith	Aspirin	75mg once daily	30
2	2	Kamal Perera	John Smith	Atorvastatin	20mg once daily	90
3	3	Nimali Silva	Sarah Lee	Ibuprofen	400mg three times daily	7
4	4	Sunil Jayawardena	Michael Brown	Paracetamol	500mg twice daily	5
5	5	Amali Fernando	Emily White	Vitamin C	500mg once daily	30

Q31)

SELECT

patients.patient_id,

CONCAT(patients.first_name, ' ', patients.last_name) **AS** *patient_name*,

COUNT(appointments.appointment_id) **AS** *total_appointments*

FROM patients

LEFT JOIN appointments **ON** patients.patient_id = appointments.patient_id

GROUP BY patients.patient_id, patients.first_name, patients.last_name

ORDER BY *total_appointments* **desc**;

patients 1 X			
SELECT patients.patient_id, Enter a SQL expression to filter results (use Ctrl+Space)			
	123 patient_id	AZ patient_name	123 total_appointments
1	1	Kamal Perera	2
2	2	Nimali Silva	2
3	3	Sunil Jayawardena	2
4	4	Amali Fernando	1
5	5	Ruwan Rathnayake	1
6	6	Dilani Bandara	0

Q32)

SELECT

wards.ward_id,

wards.ward_name,

wards.ward_type,

COUNT(nurses.nurse_id) **AS** nurse_count

FROM wards

LEFT JOIN nurses **ON** wards.ward_id = nurses.ward_id

GROUP BY wards.ward_id, wards.ward_name,wards.ward_type

ORDER BY nurse_count **DESC**;

wards 1 X					
SELECT wards.ward_id, ward Enter a SQL expression to filter results (use Ctrl+Space)					
Grid	123 ward_id	A-Z ward_name	A-Z ward_type	123 nurse_count	
1	1	Cardiac Care ICU	ICU	1	
2	3	Orthopaedic Surgical Ward	Surgical	1	
3	7	Paediatric Ward	Paediatric	1	
4	2	Cardiology General Ward	General	0	
5	4	Orthopaedic Recovery Ward	General	0	
6	5	Neurology ICU	ICU	0	
7	6	Neurology General Ward	General	0	
8	8	Neonatal ICU	ICU	0	

Q33)

SELECT

doctors.doctor_id,

CONCAT(doctors.first_name, ' ', doctors.last_name) **AS** doctor_name,

doctors.speciality,

COUNT(appointments.appointment_id) **AS** appointment_count

FROM appointments

RIGHT JOIN doctors **ON** appointments.doctor_id = doctors.doctor_id

GROUP BY doctors.doctor_id, doctors.first_name, doctors.last_name, doctors.speciality

ORDER BY appointment_count **DESC**;

doctors 1					
SELECT doctors.doctor_id, Enter a SQL expression to filter results (use Ctrl+Space)					
	123 doctor_id	A-Z doctor_name	A-Z speciality	123 appointment_count	
1	1	John Smith	Cardiologist	3	
2	2	Sarah Lee	Orthopaedic Surgeon	2	
3	3	Michael Brown	Neurologist	2	
4	4	Emily White	Paediatrician	1	
5	5	David Wilson	Cardiologist	0	

Q34)

SELECT

departments.dept_name **AS** *department_name*,

COUNT(**DISTINCT** appointments.appointment_id) **AS** *total_appointments*,

COALESCE(**SUM**(bills.totle_amount), 0) **AS** *gross_revenue*,

COALESCE(**SUM**(bills.paid_amount), 0) **AS** *collected*,

COALESCE(**SUM**(bills.totle_amount - bills.paid_amount), 0) **AS** *outstanding_balance*

FROM departments

LEFT JOIN doctors **ON** departments.dept_id = doctors.dep_id

LEFT JOIN appointments **ON** doctors.doctor_id = appointments.doctor_id

LEFT JOIN bills **ON** appointments.appointment_id = bills.appointment_id

GROUP BY departments.dept_id, departments.dept_name

ORDER BY *gross_revenue* **DESC**;

departments 1					
SELECT departments.dept_name AS department_name Enter a SQL expression to filter results (use Ctrl+Space)					
	A-Z department_name	123 total_appointments	123 gross_revenue	123 collected	123 outstanding_balance
1	Cardiology	3	7,500	7,500	0
2	Orthopaedics	2	5,000	5,000	0
3	Neurology	2	3,500	0	3,500
4	Paediatrics	1	2,500	2,500	0

Q35)

SELECT

```

CONCAT(patients.first_name, ' ', patients.last_name) AS patient_name,

patients.phone

FROM patients

WHERE patients.patient_id IN (

SELECT DISTINCT appointments.patient_id

FROM appointments

INNER JOIN bills ON appointments.appointment_id = bills.appointment_id

WHERE bills.payment_ststus IN ('Unpaid', 'Partial')

);

```

patients 1 X

SELECT CONCAT(patients.first_name, ' ', patients.la) Enter a SQL expression to filter results (use Ctrl+Spa

Grid	A-Z patient_name	A-Z phone
1	Sunil Jayawardena	0734567890
Text		

Q36)

Q37)

No	Table Name	Primary Key Column	Reason for Selection
1	departments	dept_id	To uniquely identify each department. Auto-increment provides automatic numbering when adding new departments.
2	wards	ward_id	To uniquely identify each ward. Ward names may be same across different departments.
3	doctors	doctor_id	To uniquely identify each doctor. Email is unique but can be changed over time.
4	nurses	nurse_id	To uniquely identify each nurse. Auto-increment provides convenience for identification.
5	patients	patient_id	To uniquely identify each patient. Name, phone number and other details may change over time.
6	appointments	appointment_id	To uniquely identify each appointment. Same patient and doctor can have multiple appointments.
7	prescriptions	prescription_id	To uniquely identify each prescription line item. Auto-increment integer is simple and efficient.
8	bills	bill_id	To uniquely identify each bill. Auto-increment provides automatic numbering.

Q38)

1. **wards.dept_id** references **departments.dept_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Prevents deleting a department that still has wards.
2. **doctors.dept_id** references **departments.dept_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Prevents deleting a department that still has doctors.
3. **nurses.ward_id** references **wards.ward_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Prevents deleting a ward that still has nurses.
4. **appointments.patient_id** references **patients.patient_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Preserves medical history; patients cannot be deleted if they have appointments.
5. **appointments.doctor_id** references **doctors.doctor_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Preserves medical history; doctors cannot be deleted if they have appointments.
6. **prescriptions.appointment_id** references **appointments.appointment_id** (ON DELETE CASCADE, ON UPDATE CASCADE) - When an appointment is deleted, its prescriptions are automatically deleted (they are meaningless without the appointment).
7. **bills.appointment_id** references **appointments.appointment_id** (ON DELETE RESTRICT, ON UPDATE RESTRICT) - Preserves financial records for auditing purposes; bills should remain even if appointments are deleted.

Q39)

No	Table Name	Column(s)	Constraint Name	Business Rule
1	departments	dept_name	dept_name (UNIQUE)	Same department name cannot be used twice in the hospital
2	doctors	email	email (UNIQUE)	Each doctor must have a unique email address for system login and communication
3	nurses	email	email (UNIQUE)	Each nurse must have a unique email address for system login and communication
4	prescriptions	(appointment_id, medicine_name)	unique_medicine_per_appointment	Same medicine cannot be prescribed more than once in the same appointment
5	bills	appointment_id	appointment_id (UNIQUE)	One appointment can have only one bill (one-to-one relationship)

Q40)

No	Table Name	CHECK Constraint	Plain English Description	Data Error Prevented
1	wards	capacity BETWEEN 1 AND 100	Ward capacity must be between 1 and 100 patients	Prevents entering invalid capacity (negative, zero, or over 100)
2	doctors	salary > 0	Doctor's salary must be greater than zero	Prevents entering zero or negative salary
3	prescriptions	duration_days BETWEEN 1 AND 365	Prescription duration must be between 1 and 365 days	Prevents invalid duration (zero, negative, or over 365 days)
4	bills	total_amount >= 0	Total bill amount cannot be negative	Prevents entering negative bill amounts
5	bills	paid_amount >= 0	Paid amount cannot be negative	Prevents entering negative payment amounts
6	bills	paid_amount <= total_amount	Paid amount cannot exceed total bill amount	Prevents overpayment entry errors

Q41)

ALTER TABLE prescriptions

ADD CONSTRAINT chk_duration_max_365

CHECK (duration_days <= 365);

Statistics 1 X	
Name	Value
Updated Rows	5
Execute time	0.14s
Start time	Fri May 15 15:15:42 IST 2026
Finish time	Fri May 15 15:15:42 IST 2026
Query	-- Q41
	ALTER TABLE prescriptions
	ADD CONSTRAINT chk_duration_max_365
	CHECK (duration_days <= 365)

Q42)

- "In my schema, ON DELETE RESTRICT has been used for the dep_id foreign key in the doctors table."
- If the department has doctors assigned, the department cannot be deleted and an error message will appear. If the department has no doctors, the department will be successfully deleted.
- No
- Data security and preservation of medical history

Q43)

Q44)

	PRIMARY KEY	UNIQUE
NULL values	NOT NULL (cannot be empty)	Can have NULL (one NULL allowed)
Number per table	Only one	Multiple allowed
Purpose	Uniquely identifies each row	Prevents duplicate values

```
CREATE TABLE departments(
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(100) NOT NULL UNIQUE,
    location VARCHAR(100) NOT NULL,
    phone_ext CHAR(10) NOT NULL,
    head_doctor VARCHAR(50) NULL
);
```

Q45)

- Referential integrity is a database rule that ensures relationships between tables remain consistent. It guarantees that a foreign key value in a child table always refers to an existing primary key value in the parent table.
- FOREIGN KEY (patient_id) REFERENCES patients(patient_id)

Situation	What Happens
Trying to insert a non-existent patient_id	Foreign key rejects the INSERT - Error message appears
Inserting an existing patient_id	Foreign key allows it - INSERT succeeds